# A complete classification of a primitive cubic family

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**Abstract**

We prove that a primitive polynomial $X^3+X^2+X+\lambda$ over $\mathbb{F}_{q^2}$, with $\lambda$ primitive in $\mathbb{F}_{q^2}$, exists if and only if the characteristic is not three. This determines all surviving cases of a conjecture of Awasthi and Sharma, whose case $q=3$ was recently disproved. A Kummer parametrization retains the coefficient conditions exactly and gives uniform character-sum estimates, with different main terms in odd and even characteristic. A sieve proves existence for every $q \geq 1511$, and independently verified finite-field certificates cover the remaining 261 prime powers. The proof uses the incomplete-character-sum theorem of Fu and Wan and includes an explicit accounting of its geometric hypotheses.

## 1 Introduction and statement

A monic polynomial of degree $m$ over $\mathbb{F}_Q$ is primitive if its roots generate the multiplicative group $\mathbb{F}_{Q^m}^*$. Awasthi and Sharma [1, Conjecture 9.1] asked whether there always exists a primitive polynomial

$$h_\lambda(X) = X^3 + X^2 + X + \lambda \in \mathbb{F}_{q^2}[X] \tag{1}$$

with $\lambda$ primitive in $\mathbb{F}_{q^2}$. Their surrounding discussion takes $q$ prime. Sharma [3, Section 4] disproved the case $q=3$ and considered the broader prime-power setting. We give the complete classification in that setting.

**Theorem 1.1.** *Let $q$ be any prime power. There exists a primitive element $\lambda \in \mathbb{F}_{q^2}$ for which (1) is primitive if and only if* $\operatorname{char}(\mathbb{F}_q) \neq 3$.

The positive direction is partly computer assisted: a uniform argument reduces it to a specified finite interval, and 261 explicit certificates cover that entire interval. The files `witnesses.json` and `check.py` provide the certificates and a verifier using only exact integer and polynomial arithmetic. The proof does not infer the infinite conclusion from a sample.

We also obtain a counting result. Write $P(q)$ for the number of constants in Theorem 1.1, and put

$$Q = q^2, \qquad N = Q^3 - 1 = q^6 - 1, \qquad \theta(e) = \frac{\varphi(e)}{e}, \qquad W(e) = 2^{\omega(e)},$$

where $\omega(e)$ is the number of distinct prime divisors of $e$.

**Theorem 1.2.** *If $q$ has odd characteristic different from three, then*

$$\left| \frac{3P(q)}{\theta(N)} - (q^2-1) \right| \leq 6q\big(W(N)-1\big).$$

*If $q$ is even, then*

$$\left| \frac{3P(q)}{\theta(N)} - 2q^2 \right| \leq 4q\big(W(N)-1\big).$$

*In characteristic three, $P(q) = 0$.*

Thus the even-characteristic family has twice the leading density. Since $W(n) = n^{o(1)}$, both estimates have a relative error tending to zero as $q$ tends to infinity in the respective regime.

The character estimates come from Fu and Wan [2, Proposition 2.1 and Theorem 4.2]. The Kummer parametrization below enforces both prescribed coefficients before applying those estimates. This is essential: multiplicative freeness in an extension field does not enforce membership in a smaller field. We discuss the distinction from the sufficient condition in [3] at the end.

## 2 The obstruction and a parametrization

**Lemma 2.1.** *In characteristic three, no polynomial* (1) *can be both irreducible and have primitive constant term.*

*Proof.* The discriminant of $X^3 + X^2 + X + \lambda$ in characteristic three is $-\lambda$. An irreducible cubic over an odd finite field has nonzero square discriminant: Frobenius permutes its roots as a three-cycle and therefore fixes their Vandermonde product. Since $Q = q^2$ is an odd square, $-1$ is a square in $\mathbb{F}_Q$. Consequently $\lambda$ must be a square. A generator of the even-order cyclic group $\mathbb{F}_Q^*$ is not a square. □

From now on the characteristic $p$ differs from three. Then $Q \equiv 1 \pmod 3$. Choose a noncube $a \in \mathbb{F}_Q^*$ and a root $\alpha$ of $X^3 - a$. This cubic is irreducible, since a reducible cubic has a root. Hence

$$K = \mathbb{F}_Q(\alpha) = \mathbb{F}_{Q^3}.$$

The element $\zeta = \alpha^{Q-1}$ is a primitive cube root of unity in $\mathbb{F}_Q$, and the three conjugates of $\alpha$ are $\alpha, \zeta\alpha, \zeta^2\alpha$.

For $\beta \in K$, define its subtrace by

$$\mathrm{St}(\beta) = \beta^{1+Q} + \beta^{1+Q^2} + \beta^{Q+Q^2}.$$

Every $\beta \in K$ has a unique expression $c + u\alpha + v\alpha^2$ with $c, u, v \in \mathbb{F}_Q$. Direct expansion of the three conjugates gives

$$\mathrm{Tr}_{K/\mathbb{F}_Q}(\beta) = 3c, \qquad \mathrm{St}(\beta) = 3c^2 - 3auv. \tag{2}$$

**Lemma 2.2.** *If $p \geq 5$, the elements of $K$ having trace $-1$ and subtrace $1$ are exactly*

$$b(t) = -\frac{1}{3} + \alpha t - \frac{2\alpha^2}{9at}, \qquad t \in \mathbb{F}_Q^*, \tag{3}$$

*each occurring once. If $p = 2$, they form the union of the two lines*

$$b_1(t) = 1 + \alpha t, \qquad b_2(t) = 1 + \alpha^2 t, \qquad t \in \mathbb{F}_Q, \tag{4}$$

*whose only common element is* $1$.

*Proof.* For $p \geq 5$, equations (2) give $c = -1/3$ and $uv = -2/(9a) \neq 0$. Take $u = t$ and solve for $v$. In characteristic two they give $c = 1$ and $uv = 0$. Uniqueness of the basis expansion proves both the parametrizations and the intersection assertion. □

A primitive $\beta \in K$ has degree three over $\mathbb{F}_Q$. Its minimal polynomial is

$$X^3 - \mathrm{Tr}(\beta)X^2 + \mathrm{St}(\beta)X - \mathrm{N}_{K/\mathbb{F}_Q}(\beta).$$

Moreover, the norm of a generator of $K^*$ generates $\mathbb{F}_Q^*$. Negation preserves generators of $\mathbb{F}_Q^*$ when $Q$ is even or $Q \equiv 1 \pmod 4$. In the latter case, if $z$ generates the group and $\gcd(j, Q-1) = 1$, then

$j + (Q-1)/2$ remains coprime to $Q-1$: it remains odd and is unchanged modulo every odd prime divisor. Since $Q = q^2$, one of these two cases always applies.

It follows that every primitive element in Lemma 2.2 gives a polynomial counted by $P(q)$, and every such polynomial has exactly three counted roots. In characteristic two we may count the two lines with multiplicity at 1, since 1 is not primitive in $K$.

## 3 Incomplete character sums

We record the precise specialization of the external character-sum theorem that we use. Let $f \in K(t)^*$, and let $U$ be an open subset of $\mathbb{P}^1_{\mathbb{F}_Q}$ obtained by removing the zeros and poles of all three coefficient conjugates of $f$. For a nontrivial multiplicative character $\chi$ of $K^*$, the Kummer sheaf of $(\chi, f)$ has rank one, weight zero, and only tame ramification. Fu–Wan tensor induction produces a rank-one sheaf on $U$ whose trace at $t \in U(\mathbb{F}_Q)$ is $\chi(f(t))$. After base change, its three Kummer factors have character weights $1, Q, Q^2$ on the coefficient conjugates; see [2, Proposition 2.1].

If some zero belongs to exactly one conjugate and is simple, its inertia exponent in this tensor product is one of $1, Q, Q^2$. These integers are coprime to the order of $\chi$, which divides $Q^3 - 1$. The induced sheaf is therefore geometrically nonconstant. If the removed set has $s$ geometric points, [2, Theorem 4.2], with genus zero and zero Swan conductors, gives

$$\left| \sum_{t \in U(\mathbb{F}_Q)} \chi(f(t)) \right| \le (s-2)\sqrt{Q}. \tag{5}$$

The puncture count in (5) is the actual union of the three supports. This is sharper here than counting their degrees separately. These observations check the geometric nonconstancy hypothesis; the mere nontriviality of $\chi$ would not suffice.

**Lemma 3.1.** *For every nontrivial multiplicative character $\chi$ of $K^*$, we have*

$$\left| \sum_{t \in \mathbb{F}_Q^*} \chi(b(t)) \right| \le 6\sqrt{Q} \quad (p \ge 5),$$

*and*

$$\left| \sum_{t \in \mathbb{F}_Q} \chi(b_i(t)) \right| \le 2\sqrt{Q} \quad (p = 2,\ i = 1, 2).$$

*Proof.* The two zeros of (3) are

$$\frac{2}{3\alpha}, \qquad -\frac{1}{3\alpha}.$$

Each has a Frobenius orbit of size three. The orbits could meet only if $-2$ were a cube root of unity. That would imply $-8 = 1$, impossible for $p \ge 5$. Thus the three conjugate functions have six disjoint simple zeros. Their poles are 0 and $\infty$, both simple, and there are eight punctures in total. No zero lies in $\mathbb{F}_Q$, so $U(\mathbb{F}_Q) = \mathbb{F}_Q^*$. Apply (5).

For either line in (4), its zero and the two coefficient-conjugate zeros are distinct and lie outside $\mathbb{F}_Q$. The pole is $\infty$. Hence there are four punctures and $U(\mathbb{F}_Q) = \mathbb{F}_Q$. The same nonconstancy check and (5) prove the second bound. $\square$

For a divisor $e \mid N$, call $x \in K^*$ *e-free* if it is not an $\ell$-th power for any prime $\ell \mid e$. Let $A(e)$ count $e$-free values in the parameter multiset of Lemma 2.2, counting the common point of the two lines twice in characteristic two. Every value is nonzero. The standard indicator is

$$\rho_e(x) = \theta(e) \sum_{d \mid e} \frac{\mu(d)}{\varphi(d)} \sum_{\operatorname{ord}(\chi)=d} \chi(x). \tag{6}$$

This follows by multiplying the prime-divisor indicators in the cyclic group $K^*$.

Set

$$(C, B) = \begin{cases} (Q-1, 6\sqrt{Q}), & p \geq 5, \\ (2Q, 4\sqrt{Q}), & p = 2. \end{cases} \tag{7}$$

The trivial character contributes $C$. Lemma 3.1 and (6) imply

$$\left| \frac{A(e)}{\theta(e)} - C \right| \leq \big(W(e) - 1\big) B. \tag{8}$$

Since $A(N) = 3P(q)$, this proves Theorem 1.2, including the obstruction from Lemma 2.1.

# 4 A sieve and a finite cutoff

**Lemma 4.1.** *Write* $\operatorname{rad}(N) = k\ell_1 \cdots \ell_s$, *where* $k$ *is squarefree and the* $\ell_i$ *are distinct primes not dividing* $k$. *Suppose* $s \geq 1$ *and*

$$\delta = 1 - \sum_{i=1}^{s} \frac{1}{\ell_i} > 0.$$

*Then* $A(N) > 0$ *whenever*

$$\frac{C}{B} > W(k) \left( \frac{s-1}{\delta} + 2 \right) - 1. \tag{9}$$

*For* $s = 0$, *the sufficient condition is* $C/B > W(N) - 1$.

*Proof.* Pointwise on any multiset of elements of $K^*$, the $e$-free indicators give

$$A(N) \geq \sum_{i=1}^{s} A(k\ell_i) - (s-1)A(k).$$

Indeed, an element which is not $k$-free contributes zero on the right; a $k$-free element missing at least one further prime condition contributes at most zero, and an $N$-free element contributes one.

Subtracting the character expansions for $A(k\ell_i)$ and $\theta(\ell_i)A(k)$ leaves the squarefree character orders divisible by $\ell_i$. There are $W(k)$ such orders, with normalized total weight one for each. Therefore

$$|A(k\ell_i) - \theta(\ell_i)A(k)| \leq \theta(k\ell_i)W(k)B.$$

Since $\sum_i \theta(\ell_i) = s - 1 + \delta$, inequality (8) yields

$$\begin{aligned} A(N) &\geq \delta A(k) - \theta(k)W(k)B(s-1+\delta) \\ &\geq \theta(k)\big[\delta(C+B) - W(k)B(s-1+2\delta)\big]. \end{aligned}$$

This is positive under (9). The case $s = 0$ follows directly from (8). ☐

**Proposition 4.2.** *If* $q \geq 1511$ *and* $p \neq 3$, *then* $A(N) > 0$.

*Proof.* In both cases of (7),

$$\frac{C}{B} \geq \frac{q - 1/q}{6}. \tag{10}$$

Let $t = \omega(N)$, and write $p_1 < p_2 < \cdots$ for the ordinary primes. Take $k$ to contain the $j$ smallest prime divisors of $N$. For $j < t$, put

$$d_{t,j} = 1 - \sum_{i=j+1}^{t} \frac{1}{p_i}, \qquad R_{t,j} = 2^j \left( \frac{t - j - 1}{d_{t,j}} + 2 \right) - 1,$$

using only choices with $d_{t,j} > 0$. For $j = t$, put $R_{t,t} = 2^t - 1$. The actual $\delta$ is at least $d_{t,j}$, so Lemma 4.1 applies if $q - 1/q > 6R_{t,j}$.

For $1 \leq t \leq 15$, choose respectively

$$j = 0, 1, 2, 2, 2, 2, 2, 2, 2, 2, 2, 2, 3, 3, 3.$$

Exact rational calculation gives $6R_{t,j} < 1511 - 1/1511$ in every case. For $16 \leq t \leq 56$, Table 1 gives $j$ and an integer $T$ satisfying

$$\prod_{i=1}^{t} p_i > T^6, \qquad T - 1/T > 6R_{t,j}. \tag{11}$$

Since $N = q^6 - 1$ has $t$ distinct prime factors, the first inequality implies $q > T$, and the second proves the desired sieve condition.

For the infinite tail, the exact base inequality is

$$\prod_{i=1}^{57} p_i > (6 \cdot 2^{57})^6.$$

The next prime is $271 > 64$, as are all subsequent primes. Induction therefore proves $\prod_{i=1}^{t} p_i > (6 \cdot 2^t)^6$ for every $t \geq 57$. Hence $q > 6 \cdot 2^t$, which implies $q - 1/q > 6(2^t - 1)$. The direct case $j = t$ suffices.

All finite rational inequalities, prime products, and the tail base inequality are checked by the supplied verifier. This completes the cutoff proof. □

# 5 Finite certificates and completion of the proof

For each of the 261 prime powers $q = p^r < 1511$ with $p \neq 3$, the file `witnesses.json` supplies a monic polynomial $M(Z)$ of degree $2r$ over $\mathbb{F}_p$ and an exponent $v$. If $z$ denotes the residue class of $Z$, the certificate verifies that $z$ has exact order $Q - 1$ in $\mathbb{F}_p[Z]/(M)$. This proves that the quotient is a field: the cyclic subgroup generated by $z$ already contains all $Q - 1$ nonzero residue classes. It is therefore a specified model of $\mathbb{F}_Q$.

The exponent satisfies $\gcd(v, Q - 1) = 1$, and $\lambda = z^v$. In the quotient by $h_\lambda$, the residue class $x$ of $X$ is checked to satisfy

$$x^N = 1, \qquad x^{N/\ell} \neq 1 \quad \text{for every prime } \ell \mid N. \tag{12}$$

Thus its order is exactly $N$. Independently, the verifier checks

$$\gcd(X^Q - X, h_\lambda) = 1.$$

| $t$ | $j$ | $T$ | $t$ | $j$ | $T$ | $t$ | $j$ | $T$ |
|---|---|---|---|---|---|---|---|---|
| 16 | 3 | 1724 | 30 | 3 | 6891 | 44 | 4 | 15968 |
| 17 | 3 | 1949 | 31 | 3 | 7469 | 45 | 4 | 16726 |
| 18 | 3 | 2195 | 32 | 3 | 8090 | 46 | 4 | 17515 |
| 19 | 3 | 2456 | 33 | 3 | 8752 | 47 | 4 | 18316 |
| 20 | 3 | 2736 | 34 | 3 | 9470 | 48 | 4 | 19129 |
| 21 | 3 | 3040 | 35 | 4 | 10129 | 49 | 4 | 19969 |
| 22 | 3 | 3363 | 36 | 4 | 10701 | 50 | 4 | 20841 |
| 23 | 3 | 3710 | 37 | 4 | 11290 | 51 | 4 | 21743 |
| 24 | 3 | 4078 | 38 | 4 | 11897 | 52 | 4 | 22671 |
| 25 | 3 | 4464 | 39 | 4 | 12525 | 53 | 4 | 23634 |
| 26 | 3 | 4878 | 40 | 4 | 13171 | 54 | 4 | 24617 |
| 27 | 3 | 5326 | 41 | 4 | 13837 | 55 | 4 | 25630 |
| 28 | 3 | 5807 | 42 | 4 | 14529 | 56 | 4 | 26673 |
| 29 | 3 | 6329 | 43 | 4 | 15235 | | | |

Table 1: Exact finite sieve certificates for (11).

A cubic with no root in $\mathbb{F}_Q$ is irreducible, so this also verifies the field property of the cubic quotient directly. Consequently each certificate gives a primitive polynomial of the required form.

There is no unverified factorization in (12). The full prime support of $N$ is obtained by factoring the four integers in

$$q^6 - 1 = (q-1)(q+1)(q^2+q+1)(q^2-q+1).$$

For this finite interval, each factor is less than $1511^2 + 1511 + 1$. Trial division is enough, and the verifier removes all resulting prime powers from $N$ and checks that the residual factor is one. It likewise verifies the complete set of prime powers below the cutoff, rather than merely reading a claimed count.

The witness generator uses C++ integer encodings of finite-field elements. The separate Python verifier uses coefficient tuples and independent polynomial arithmetic. It verifies the base-field generator, the primitive constant, cubic irreducibility, all root-order tests, the 261-case coverage, and every finite inequality in Proposition 4.2. Neither implementation uses floating-point arithmetic for a mathematical decision.

*Proof of Theorem 1.1.* Lemma 2.1 excludes characteristic three. In all other characteristics, Proposition 4.2 proves existence when $q \geq 1511$. The finite certificates prove existence for every remaining prime power $q$. The norm argument after Lemma 2.2 ensures that all counted constants are primitive. These cases exhaust the theorem. □

## 6 Relation to earlier claims and further questions

The case $q = 3$ was already disproved in [3]; we do not claim priority for that counterexample. The new conclusion is the complete classification of the surviving cases, together with the two uniform counting estimates. The general degree-$m$ existence problem for polynomials $g(X)+\lambda$, with nonconstant coefficients in a subfield, remains outside the theorem.

The sufficient-condition argument in [3, Section 5] counts pairs in which $-g(\beta)$ is $(q^n-1)$-free in $\mathbb{F}^*_{q^{mn}}$. That condition does not require $-g(\beta)$ to lie in $\mathbb{F}_{q^n}$, so it is not equivalent to the stated primitive-constant condition. Its general conclusion also conflicts with a norm obstruction: if $m$ is odd and $q^n \equiv 3 \pmod 4$, the norm of a primitive root would be $-\lambda$, while a primitive $\lambda$ has nonprimitive negative. Taking $m = n = 3$ and arbitrarily large $q = 7^{2r+1}$ meets the characteristic hypothesis of its Theorem 5.1 and exhibits this incompatibility. Our proof for the cubic family uses no such equivalence.

The parametrization gives a more general route for prescribed-coefficient questions: impose the trace equations first, then check that the induced multiplicative characters are geometrically nonconstant on the resulting curve. The characteristic-three obstruction here shows that this last condition cannot be omitted. Extending the classification to other pairs of prescribed cubic coefficients, or to higher-degree subfield families, requires a new analysis of those curves and their exceptional characters.

## References


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